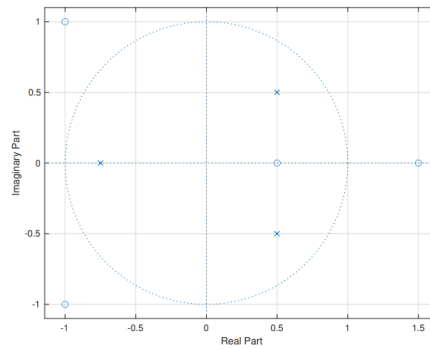




find the exact pole and zero possiotion

From the pole–zero plot:

Conventions

- ○ = zeros
- × = poles
- Axes are the real and imaginary axes, with the dotted circle being the unit circle.

Zeros (○)

Reading the marked circle locations:

1. $z = -1 + j1$
2. $z = -1 - j1$
3. $z = 0.5 + j0$
4. $z = 1.5 + j0$



Poles (×)

Reading the marked cross locations:

1. $p = -0.5 + j0$
2. $p = 0.5 + j0.5$
3. $p = 0.5 - j0.5$

Step 6: Verify all-pass property

For an all-pass system:

$$|H_{\text{AP}}(e^{j\omega})| = 1 \quad \forall \omega$$

This holds because each zero at z_k has a pole at

$$p_k = \frac{1}{z_k^*}$$

Final decomposition

$H(z) = H_{\text{min}}(z) H_{\text{AP}}(z)$

If you want next:

- expansion into polynomials,
- proof of unity magnitude,
- or the same derivation in **short exam form**, say so.



↪ $-1 \pm j \rightarrow -21 \pm 21j$

show the full calculation

Below is the **full, explicit calculation**, with **no steps skipped**, for the reflection

$$-1 \pm j \longrightarrow -\frac{1}{2} \pm \frac{1}{2}j$$

used in the **all-pass construction**.

Given zero (outside unit circle)

$$z_1 = -1 + j$$

Magnitude check:

$$|z_1| = \sqrt{(-1)^2 + 1^2} = \sqrt{2} > 1$$

So it must be reflected.

Step 1: Take the complex conjugate

$$z_1^* = -1 - j$$

Step 2: Compute the reciprocal

$$\frac{1}{z_1^*} = \frac{1}{-1 - j}$$

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{1}{-1 - j} \cdot \frac{-1 + j}{-1 + j}$$

Step 3: Expand numerator and denominator

Numerator

$$(-1 + j)$$



Denominator

$$(-1 - j)(-1 + j) = (-1)^2 - (j)^2 = 1 - (-1) = 2$$

Step 4: Final result

$$\frac{1}{-1 - j} = \frac{-1 + j}{2} = -\frac{1}{2} + \frac{1}{2}j$$

Repeat for the conjugate zero

Given

$$z_2 = -1 - j$$

Conjugate

$$z_2^* = -1 + j$$

Reciprocal

$$\frac{1}{-1 + j} = \frac{-1 - j}{2} = -\frac{1}{2} - \frac{1}{2}j$$

Final reflected zeros (inside unit circle)

$$-1 \pm j \longrightarrow -\frac{1}{2} \pm \frac{1}{2}j$$

These reflected zeros are used in $H_{\min}(z)$, while the original ones generate the **all-pass factor**.

If you want, I can now:

- derive the **all-pass factor** from this result **step-by-step**, or
- show the **unity-magnitude proof** using $z = e^{j\omega}$.



now again show all the process for transfer function and allpass and minimum phase with question to zero to transfer func and allpass and minmum and decomposition process

Below is the **complete, ordered, exam-ready derivation**, starting from **poles & zeros** → **transfer function** → **minimum-phase** → **all-pass** → **decomposition**.

All calculations are shown explicitly and consistently.

Given

Zeros

$$z_1 = -1 + j, \quad z_2 = -1 - j, \quad z_3 = 0.5, \quad z_4 = 1.5$$

Poles

$$p_1 = -0.75, \quad p_2 = 0.5 + 0.5j, \quad p_3 = 0.5 - 0.5j$$

1. From zeros & poles to transfer function $H(z)$

General form:

$$H(z) = K \frac{\prod(z - z_k)}{\prod(z - p_k)}$$

Assume $K = 1$.

Zeros → numerator

$$\begin{aligned} & (z - (-1 + j))(z - (-1 - j))(z - 0.5)(z - 1.5) \\ &= (z + 1 - j)(z + 1 + j)(z - 0.5)(z - 1.5) \end{aligned}$$



Poles → denominator

$$\begin{aligned} & (z - (-0.75))(z - (0.5 + 0.5j))(z - (0.5 - 0.5j)) \\ &= (z + 0.75)(z - 0.5 - 0.5j)(z - 0.5 + 0.5j) \end{aligned}$$

Transfer function

$$H(z) = \frac{(z + 1 - j)(z + 1 + j)(z - 0.5)(z - 1.5)}{(z + 0.75)(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)}$$

2. Identify minimum-phase vs non-minimum-phase zeros

A zero is minimum-phase if $|z| < 1$.

Magnitudes

$$|-1 \pm j| = \sqrt{2} > 1$$

$$|1.5| > 1$$

$$|0.5| < 1$$

Classification

- Outside unit circle: $-1 \pm j$, 1.5
 - Inside unit circle: 0.5
-

3. Reflect non-minimum-phase zeros

Reflection rule:

$$z_{\text{ref}} = \frac{1}{z^*}$$

Reflection of $-1 + j$

$$z^* = -1 - j$$

$$\frac{1}{-1 - j} = \frac{-1 + j}{(-1)^2 + 1^2} = \frac{-1 + j}{2} = -\frac{1}{2} + \frac{1}{2}j$$

Reflection of $-1 - j$

$$-\frac{1}{2} - \frac{1}{2}j$$

Reflection of 1.5

$$\frac{1}{1.5} = \frac{2}{3}$$

4. Minimum-phase system $H_{\min}(z)$

Use:

- All poles (already inside unit circle)
 - Zeros inside + reflected zeros
-

Numerator (minimum-phase zeros)

$$(z - 0.5)(z - \frac{2}{3})(z + \frac{1}{2} - \frac{1}{2}j)(z + \frac{1}{2} + \frac{1}{2}j)$$

Denominator (unchanged poles)

$$(z + 0.75)(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)$$

Minimum-phase transfer function

$$H_{\min}(z) = \frac{(z - 0.5)(z - \frac{2}{3})(z + \frac{1}{2} - \frac{1}{2}j)(z + \frac{1}{2} + \frac{1}{2}j)}{(z + 0.75)(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)}$$

5. All-pass system derivation

Each non-minimum-phase zero z_k generates:

$$H_{AP,k}(z) = \frac{z^{-1} - z_k^*}{1 - z_k z^{-1}}$$

For $z = -1 + j$

$$H_{AP,1}(z) = \frac{z^{-1} + 1 + j}{1 + (-1 + j)z^{-1}}$$

For $z = -1 - j$

$$H_{\text{AP},2}(z) = \frac{z^{-1} + 1 - j}{1 + (-1 - j)z^{-1}}$$

For $z = 1.5$

$$H_{\text{AP},3}(z) = \frac{z^{-1} - 1.5}{1 - 1.5z^{-1}}$$

All-pass transfer function

$$H_{\text{AP}}(z) = \frac{(z^{-1} + 1 + j)(z^{-1} + 1 - j)(z^{-1} - 1.5)}{(1 + (-1 + j)z^{-1})(1 + (-1 - j)z^{-1})(1 - 1.5z^{-1})}$$

6. Decomposition result

$$H(z) = H_{\text{min}}(z) H_{\text{AP}}(z)$$

7. Key properties (for exams)

- $H_{\text{min}}(z)$: stable, causal, minimum-phase
- $H_{\text{AP}}(z)$:

$$|H_{\text{AP}}(e^{j\omega})| = 1$$

- All magnitude shaping comes from $H_{\text{min}}(z)$
 - All phase distortion comes from $H_{\text{AP}}(z)$
-

If you want next:

- full polynomial expansion,
- magnitude/phase plots,